

# PAPER\_749: Five Quantum Variable Document Sets — $r_j$ , $d_g$ , $F_U$ , $f_{\text{feedback}}$ , $\Omega_g$ , $f_{\text{Heaviside}}$ , $H_{\text{SCm}}$ , $_{i}$ , $M_{\text{bh}}$ , $_{j}$ , $E_{\text{react}}$

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## Abstract

Three sets of five quantum variable documents (15 variables total) were assimilated into the UQFF knowledge base during the Compression Cycle 2 thread. This paper consolidates all 15 variables with their equations, canonical values, and roles within the Unified Field Strength  $F_U$  formula. The variables span spatial distances ( $r_j$ ,  $d_g$ ), field strengths ( $F_U$ ,  $_{j}$ ), dynamics ( $\Omega_g$ ,  $f_{\text{feedback}}$ ,  $_{i}$ ), and operational parameters ( $f_{\text{Heaviside}}$ ,  $H_{\text{SCm}}$ ,  $M_{\text{bh}}$ ,  $E_{\text{react}}$ ). Together they define the complete parameterization of the UQFF for galactic-scale applications.

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## 1. Introduction

Multiple document sets uploaded to the Grok UQFF thread on May 08, 2025 comprised 15 individual quantum variable definition documents, each providing:

- Variable symbol and definition
- Value and units
- Role in  $U_m$ ,  $U_{gi}$ ,  $U_{bi}$ ,  $F_U$  equations
- Example calculation for the Sun at  $t=0$ ,  $t_n=0$

This paper assimilates all 15 variables as a unified reference set.

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## 2. Set A: Spatial and Field Variables ( $r_j$ , $d_g$ , $F_U$ , $f_{\text{feedback}}$ , $\Omega_g$ )

$r_j$  — Magnetic String Distance

$$r_j = 1.496 \times 10^{13} \text{ m} = 100 \text{ AU}$$

Role: Distance along  $j$ -th magnetic string path (denominator in  $U_m$  and  $U_{g3}$ )

$$U_m: \text{ }_j/r_j \rightarrow U_m \quad 2.28 \times 10 \quad \text{J/m}^3$$

$$U_{g3}: k_3 \cdot \Sigma_j B_j \cdot \cos(\theta_s \cdot t) \cdot P_{\text{core}} \cdot E_{\text{react}} \quad 1.8 \times 10^3 \text{ J/m}^3$$

**d\_g — Galactic Center Distance**

$$d_g = 2.55 \times 10^2 \text{ m} \quad 27,000 \text{ light-years}$$

Role: Distance from Sun to Milky Way center (Sgr A\* reference)

$$U_{bi}: -\rho_i \cdot U_{gi} \cdot \Omega_g \cdot (M_{bh}/d_g) \cdot (1 + \rho_{sw} \cdot \rho_{vac,sw}) \cdot U_{UA} \cdot \cos(\theta \cdot t_n)$$

$$M_{bh}/d_g = 8.15 \times 10^3 / 2.55 \times 10^2 \quad 3.20 \times 10^1 \text{ kg/m}$$

$$U_{b1} \quad -1.94 \times 10^2 \text{ J/m}^3$$

$$U_{g4}: k_4 \cdot \rho_{vac, [SCm]} \cdot (M_{bh}/d_g) \cdot e^{(-t)} \cdot \cos(\theta \cdot t_n) \cdot (1 + f_{\text{feedback}})$$

$$U_{g4} \quad 2.50 \times 10^2 \text{ J/m}^3$$

**F\_U — Unified Field Strength**

$$F_U = \Sigma_i [k_i \cdot U_{gi} - \rho_i \cdot U_{gi} \cdot \Omega_g \cdot (M_{bh}/d_g) \cdot E_{\text{react}}]$$

$$+ \Sigma_j [\rho_j / r_j \cdot (1 - e^{(-t)} \cdot \cos(\theta \cdot t_n)) \cdot \rho_j]$$

$$+ (g_{\text{un}} + \rho \cdot T_s^{(n)})$$

$$- \Sigma_i [\rho_i \cdot U_{gi} \cdot E_{\text{react}}]$$

$$\text{At } t=0, \text{ Sun: } F_U \quad U_m \quad 2.28 \times 10^3 \text{ J/m}^3 \text{ (} U_m \text{ dominates)}$$

**f\_feedback — AGN Feedback Factor**

$$f_{\text{feedback}} = 0.1 \quad (\text{for } \Delta MBH = 1 \text{ dex AGN feedback})$$

Role: Scales AGN feedback in U\_g4

$$\text{With } f_{\text{feedback}} = 0.1: U_{g4} \quad 2.50 \times 10^2 \text{ J/m}^3$$

$$\text{Without (} f_{\text{feedback}} = 0): U_{g4} \quad 2.27 \times 10^2 \text{ J/m}^3$$

Feedback effect: ~10% increase → important for galaxy evolution modeling

**$\Omega_g$  — Galactic Spin Rate**

$$\Omega_g = 7.3 \times 10^{-1} \text{ rad/s}$$

Role: Milky Way angular velocity (appears in U\_bi buoyancy term)

$$\text{Rotational period: } T = 2 / \Omega_g \quad 8.61 \times 10^1 \text{ s} \quad 2.73 \times 10 \text{ yr (galactic year)}$$

### 3. Set B: Operational Parameters (f\_Heaviside, i, H\_SCm, \_\_i, j)

#### f\_Heaviside — Heaviside Component Fraction

$$f_{\text{Heaviside}} = 0.01$$

Role: Scales threshold-activated nonlinear effects in  $U_m$

$$\text{Effect in } U_m: (1 + 10^{1.3} \cdot f_{\text{Heaviside}}) = (1 + 10^{1.1})$$

This amplifies  $U_m$  by factor  $\sim 10^{1.1}$

$$\text{Without } f_{\text{Heaviside}}: U_m \quad 2.28 \times 10^{-3} \text{ J/m}^3$$

$$\text{With } f_{\text{Heaviside}}: U_m \quad 2.28 \times 10^{-3} \text{ J/m}^3 \quad \leftarrow \text{canonical value}$$

#### i — Gravity Index

$$i \in \{1, 2, 3, 4\} \quad (\text{integer index for } U_{g1}, U_{g2}, U_{g3}, U_{g4})$$

Role: Indexes the 4 universal gravity components in  $F_U$  summation

$$\Sigma(k_i \cdot U_{gi}) = k_1 \cdot U_{g1} + k_2 \cdot U_{g2} + k_3 \cdot U_{g3} + k_4 \cdot U_{g4}$$

$$\text{At } t=0, \text{ Sun: } 1.42 \times 10^{-3} \text{ J/m}^3 \quad (U_{g2} \text{ dominant})$$

#### H\_SCm — Heliosphere Thickness Factor

$$H_{\text{SCm}} \sim 1 \quad (\text{dimensionless})$$

Role: Scales heliospheric thickness in  $U_{g2}$

$$U_{g2} = k_2 \cdot (\_vac, [UA] + \_vac, [SCm]) \cdot M_s / r^2 \cdot S(r - R_b) \cdot (1 + \_sw \cdot v_{sw}) \cdot H_{\text{SCm}} \cdot E_{\text{react}}$$

$$\text{With } H_{\text{SCm}} = 1.0: U_{g2} \quad 1.18 \times 10^{-3} \text{ J/m}^3$$

$$\text{With } H_{\text{SCm}} = 1.1: U_{g2} \quad 1.30 \times 10^{-3} \text{ J/m}^3 \quad (+10\% \text{ heliosphere thickening})$$

#### \_\_i — Inertia Coupling Constant

$$\_i = 1.0 \quad (\text{uniform for all } i)$$

Role: Scales Universal Inertia  $U_i$  in  $F_U$

$$U_i = \_i \cdot \_vac, [SCm] \cdot \_vac, [UA] \cdot \_s(t) \cdot \cos(\cdot t_n) \cdot (1 + f_{\text{TRZ}})$$

$$= 1.0 \times 7.09 \times 10^{-3} \times 7.09 \times 10^{-3} \times 2.5 \times 10^{-3} \times 1 \times 1.1$$

$$1.38 \times 10^{-3} \text{ J/m}^3$$

$$\text{Net contribution: } -\_i \cdot U_i \cdot E_{\text{react}} \quad -0.138 \text{ J/m}^3$$

#### **j — Magnetic String Index**

j = integer index for magnetic strings in U\_m and U\_g3

Role: Indexes individual magnetic field strings

$\Sigma_j$  acts over all contributing magnetic strings at the field point

At solar scale: single dominant string (j=1)

At galactic scale: multiple strings possible

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#### **4. Set C: Dynamical Variables (M\_bh, \_j, P\_core, t\_n, ) and ( , E\_react, f\_quasi, R\_b)**

**M\_bh — Black Hole Mass (Sgr A\*)**

$$M_{bh} = 8.15 \times 10^3 \text{ kg} \quad 4.1 \times 10 \text{ M}$$

Role: Sgr A\* mass scaling galactic gravitational field

Appears in U\_bi and U\_g4 as M\_bh/d\_g ratio

**\_j — Magnetic Moment (time-dependent)**

$$_j(t) = (10^3 + 0.4 \cdot \sin(_c \cdot t)) \cdot 3.38 \times 10^2 \text{ T} \cdot \text{pm}^3$$

$$_c = 2 / (3.96 \times 10 \text{ s}) \text{ (solar magnetic cycle frequency)}$$

$$\text{At } t=0: \quad _j = 10^3 \times 3.38 \times 10^2 = 3.38 \times 10^{23} \text{ T} \cdot \text{pm}^3$$

$$\text{At } t=1000 \text{ days: } (1 - e^{-(t)} \cdot \cos(_c \cdot t_n)) \quad 0.049 \rightarrow U_m \text{ scales accordingly}$$

**— Reciprocation Decay Rate**

$$= 5 \times 10 \text{ day}^{-1}$$

Role: Controls temporal decay of magnetic string effects in U\_m

$$1 - e^{-(t)} \rightarrow \text{small for } t \ll 1 / 20,000 \text{ days}$$

$$\text{At } t=1000 \text{ days: } 1 - e^{-(0.05)} \quad 0.049 \text{ (still growing)}$$

**E\_react — Reactor Efficiency Factor**

$$E_{react} = 10$$

Role: Universal scaling factor in all U\_gi and U\_m terms

Relates UQFF energy densities to physical observables

This constant is the primary bridge between the  
 $\_vac, [SCm]$  density scale ( $\sim 10^{-3}$ ) and classical physics scales.

#### **f\_quasi — Quasi-Longitudinal Wave Factor**

$f\_quasi = 0.01$

Role: Scales quasi-longitudinal wave contribution in  $U\_m$

$(1 + f\_quasi) = 1.01$  (1% correction to  $U\_m$ )

Models standing waves that form quasi-longitudinal components  
in plasma environments, relevant to Red Dwarf Reactor dynamics.

#### **R\_b — Radius of Outer Field Bubble**

$R\_b = 1.496 \times 10^{13} \text{ m} = 100 \text{ AU}$  (heliospheric termination shock)

Role: Step function boundary in  $U\_g2$

$S(r - R\_b) = 1$  for  $r \geq R\_b$  (heliosphere active)  
 $= 0$  for  $r < R\_b$  (interior region, different physics)

This defines the aether-superconductive boundary layer.

#### **P\_core — Planetary Core Penetration Factor**

$P\_core \sim 1.0$  (Sun, stars)

$P\_core \sim 10^{-3}$  (planets, moons)

Role: Scales magnetic string core penetration in  $U\_g3$

$U\_g3(\text{Sun}) \quad 1.8 \times 10^{-10} \text{ J/m}^3$

$U\_g3(\text{planet}) \quad 1.8 \times 10^{-13} \text{ J/m}^3$  (3 orders lower)

#### **t\_n — Negative Time Factor**

$t\_n = t - t_0$  (allows  $t\_n < 0$ )

Role: Time reference in oscillatory terms  $\cos(\cdot t\_n)$

For  $t = 1000$  days,  $t\_n = -1$ :

$\cos(\cdot(-1)) = -1$  (phase reversal)

$U\_gi \rightarrow \text{negative} \rightarrow \text{system in negentropic regime}$

## 5. Unified Field Strength — Complete Parameterized Equation

With all 15 variables defined:

$$\begin{aligned}
 F_U = & \sum_{i=1}^4 [k_i \cdot U_{gi} - \omega_i \cdot U_{gi} \cdot \Omega_g \cdot (M_{bh}/d_g) \cdot E_{react}] \\
 & + \sum_j [ \omega_j(t)/r_j \cdot (1 - e^{-(t)} \cdot \cos(\omega_j \cdot t_n)) \cdot \omega_j ] \\
 & \cdot P_{SCm} \cdot E_{react} \cdot (1 + 10^{13} \cdot f_{Heaviside}) \cdot (1 + f_{quasi}) \\
 & + H_{SCm} \cdot (g_{\omega} + \omega_j \cdot T_s^{(\omega_j)}) \\
 & - \omega_i \cdot \sum_i [U_i \cdot E_{react}]
 \end{aligned}$$

Where all symbols are defined by the 15 quantum variable documents above.

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## 6. Conclusion

The 15 quantum variable documents from the thread\_06Jun2025.txt provide the complete parameterization of the UQFF unified field strength  $F_U$ . Each variable has been confirmed with numerical values, equations, and solar-scale calculations. Together they establish a fully specified, quantitative framework enabling computation of  $F_U$  for any astrophysical system given mass, distance, magnetic field, and temporal parameters.

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